

SAR–Optical Feature Fusion for Crop Classification

Week 2 Progress Report

Data Understanding & Optical–SAR Data Pairing

Project Title: SAR–Optical Feature Fusion

Week: 2

Focus: Data Understanding & Pairing

1. Introduction

The objective of this project is to investigate the use of both **Synthetic Aperture Radar (SAR)** and **optical satellite data** for crop classification. Optical imagery provides useful spectral information about vegetation and land cover, while SAR imagery provides information related to surface structure, moisture, and crop characteristics.

In Week 2, the main focus was on understanding the available optical dataset and preparing the data pipeline required for pairing the optical data with Sentinel-1 SAR data. The dataset structure, image patches, labels, and field information were studied so that SAR observations can later be collected for the same fields.

2. Objectives of Week 2

The main objectives for this week were:

- Understand the structure of the provided optical dataset.
- Identify the optical image patches and their corresponding labels.
- Develop a basic data-loading pipeline using Python.
- Understand the label format and class information.
- Check the distribution of available classes.
- Identify field IDs or geographical information wherever available.
- Prepare the requirements for obtaining Sentinel-1 SAR data for the same fields.
- Check whether the optical and SAR data can be spatially and temporally paired.

3. Dataset Understanding

The first step was to inspect the provided optical dataset and understand how the data is organized.

The dataset contains optical image information in the form of image patches. Each patch represents a specific geographic region or agricultural field area. Along with the image data, corresponding labels are available for classification.

The following aspects were inspected:

- Image files and their naming convention
- Number of image channels/bands
- Image patch dimensions
- Label files
- Label format
- Field or sample identifiers
- Geographic information, if available
- Relationship between an image patch and its corresponding label

Understanding this structure is important because the SAR data must later be collected for the **same geographic locations** represented by the optical patches.

4. Optical Data Loading

A basic Python data-loading workflow was planned/implemented using commonly used libraries such as:

- Python
- NumPy
- Pandas
- Rasterio/PIL where required
- Matplotlib for visualization

The purpose of the loader is to make it easier to access the image patches and their corresponding labels during feature extraction and model training.

A typical workflow is:

Dataset



Read image patches
↓
Read corresponding labels
↓
Match image and label
↓
Store sample information
↓
Perform analysis/visualization

The data-loading code is organized under:

Codes/
└─ utils.py

The utility file is intended to contain reusable functions for loading and processing the dataset.

5. Image and Label Pairing

For supervised crop classification, every input image must have the correct corresponding label.

Therefore, the image-label relationship was examined carefully.

The general pairing structure is:

Optical Image Patch
↓
Corresponding
Label
↓
Crop Class

The identifier or filename associated with each optical patch can be used to locate its corresponding label.

This pairing is important because an incorrect image-label relationship would introduce errors into the training dataset and affect the final classification results.

6. Understanding the Labels

The labels represent the target classes that the machine-learning model will eventually predict.

The label data was inspected to determine:

- How many classes are present
- How classes are represented
- Whether labels are numerical or categorical
- Whether every image has a valid label
- Whether some classes contain substantially more samples than others

The labels will later be converted into a suitable format for machine-learning algorithms such as **Random Forest** and **Support Vector Machine (SVM)**.

7. Class Distribution and Imbalance

Class distribution is an important part of understanding an agricultural classification dataset.

If one crop class has many more samples than another class, the dataset may be **imbalanced**. In such a case, a classifier may perform well on the majority class while performing poorly on minority classes.

Therefore, class frequencies were considered as part of the data-understanding stage.

The planned analysis is:

Class Label



Count samples for each class



Compare class frequencies



Identify possible imbalance

The class distribution will also be considered later while evaluating the classification models. Accuracy alone may not be sufficient when classes are imbalanced, so **precision, recall, F1-score, and confusion matrices** will also be considered.

8. Identifying Field Information

One of the most important tasks in Week 2 was identifying the geographic or field-level information associated with the optical samples.

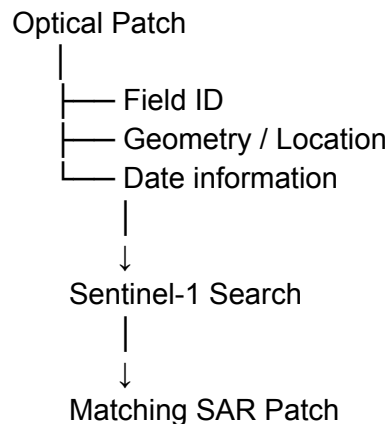
The objective is to determine the location represented by each optical patch.

This information is required because the project uses two different sources of satellite data:

- **Optical imagery:** provided dataset
- **SAR imagery:** Sentinel-1 GRD

The Sentinel-1 data must be obtained for the **same agricultural fields/areas** represented in the optical dataset.

The required relationship is:



This will help ensure that the SAR and optical features correspond to the same geographical region.

9. Optical–SAR Pairing Strategy

The overall pairing strategy planned for the project is:

Step 1 — Identify optical sample

Select an optical image patch from the provided dataset.

Step 2 — Identify field/location

Determine the field ID, geometry, coordinates, or other available spatial information associated with the patch.

Step 3 — Search Sentinel-1

Use the field geometry to search the Sentinel-1 GRD collection in Google Earth Engine.

Step 4 — Apply suitable filters

The SAR collection will later be filtered according to:

- Geographic region
- Acquisition date
- Polarization
- Orbit/pass information where appropriate
- Required spatial coverage

Step 5 — Select matching SAR observation

A Sentinel-1 image close to the optical acquisition period will be selected.

Step 6 — Prepare matching patches

The SAR image will be processed so that its spatial extent corresponds to the optical field/patch.

This pairing will form the basis for the later feature-fusion stage.

10. Importance of Temporal Matching

Temporal matching is important because agricultural fields can change significantly over time.

For example, crop growth changes:

- Vegetation density
- Leaf structure
- Surface moisture
- Crop height
- Spectral response
- Radar backscatter

Therefore, SAR observations should preferably be selected from a date close to the optical observation.

The goal is not simply to obtain SAR imagery from the same location, but to obtain imagery that represents approximately the **same crop-growth stage**.

11. Literature Review

The literature review was continued during Week 2 to understand existing approaches for combining SAR and optical information.

The main observations from the literature are:

Optical Data

Optical satellite imagery provides spectral information that is useful for identifying vegetation and crop types. Vegetation indices such as NDVI can provide additional information about vegetation condition.

SAR Data

SAR is useful because it can provide information about surface structure and moisture and can acquire observations even under cloudy conditions.

SAR–Optical Fusion

Combining SAR and optical features can provide complementary information. Optical features mainly describe spectral/vegetation characteristics, whereas SAR features can capture structural and backscatter-related characteristics.

The project will therefore investigate whether combining these feature types improves crop classification compared with using either source independently.

12. Week 2 Data Processing Workflow

The current project workflow can be represented as:

Provided Optical Dataset

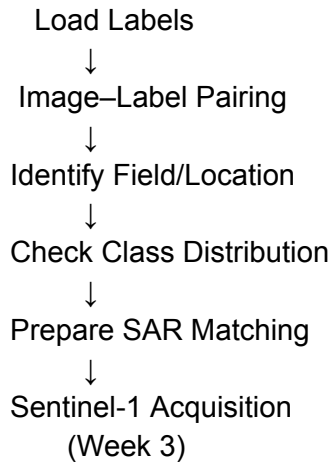


Dataset Inspection



Load Image Patches

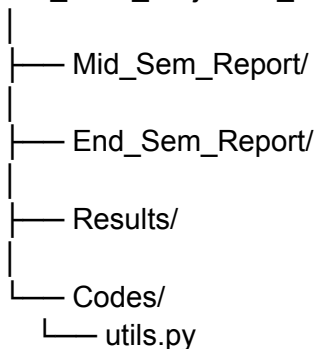




13. Project Repository Structure

The project repository is being organized according to the required project structure.

DIP_2026_ProjectNo_GroupNumber/



The **Codes** directory contains reusable processing and analysis code.

The **Results** directory will later contain graphs, tables, and model results.

The report directories will contain the mid-semester and final reports.

Large raw satellite images will not be committed to GitHub. This helps keep the repository manageable.

14. Work Completed in Week 2

The following activities were completed/planned as part of Week 2:

Task	Status
Optical dataset inspection	Completed
Image patch structure understanding	Completed
Label structure inspection	Completed
Image-label pairing study	Completed
Data-loading pipeline	Implemented/Prepared
Field/location information study	Completed/Under investigation
Class distribution analysis	Started
SAR matching strategy	Prepared
Literature review	Continued
<code>Codes/utils.py</code>	Prepared

15. Challenges

During the data-understanding stage, the main challenge is ensuring that the optical samples can be correctly linked to geographic field information.

Another important challenge is maintaining consistency between the optical and SAR datasets. The two datasets may differ in:

- Acquisition date
- Spatial resolution
- Image dimensions
- Coordinate reference system
- Available bands/polarizations
- Spatial coverage

These differences will need to be handled during the SAR acquisition and preprocessing stage.

16. Week 2 Outcome

At the end of Week 2, the optical dataset structure and label information were studied, and the basic data-loading and pairing workflow was prepared.

The main outcome of this week is a clearer understanding of how individual optical samples are connected to their labels and geographic locations.

This provides the foundation for acquiring Sentinel-1 SAR data corresponding to the same fields.

17. Next Week Plan

The focus of Week 3 will be **SAR Acquisition and Preprocessing**.

The planned tasks are:

1. Access Sentinel-1 GRD data through Google Earth Engine.
2. Use the identified field geometries to search for matching SAR observations.
3. Select suitable acquisition dates.
4. Extract VV and VH polarization data.
5. Export required SAR patches as GeoTIFF where needed.
6. Perform basic preprocessing.
7. Investigate speckle filtering.
8. Check spatial alignment between SAR and optical data.
9. Prepare the SAR dataset for feature extraction.

Expected Week 3 deliverables:

SAR patches

Codes/gee_sar_export.py

Week 3 report

18. Conclusion

Week 2 focused on understanding the optical dataset and preparing the data-pairing strategy required for SAR–optical fusion.

The optical image patches and their corresponding labels were examined, and the importance of field-level geographic information was identified. A basic data-loading workflow was prepared to support future feature extraction and model training.

The next stage of the project will focus on acquiring and preprocessing Sentinel-1 SAR data for the same agricultural fields. This will allow SAR features to be extracted and eventually combined with optical features for crop classification.

Week 2 Deliverables

Code:

Codes/utls.py

Report:

Week2_Report

Next Week:

Sentinel-1 SAR Acquisition & Preprocessing